Mutual Hand–Object Reasoning
for Egocentric Manipulation Perception

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Abstract

Identifying the object a person is manipulating, and which hand is doing it, matters for understanding demonstrations. Existing egocentric hand–object segmentation methods mostly let the hand identify the object. That hand-to-object path fails when the hand is occluded or only partly visible. We treat the hand and the manipulated object as mutually informative, and structurally different. The model first grounds *where* the interaction is, using hand, object, and contact priors. It then represents *what* is involved with two identity-stable hand queries and a set of object hypotheses, and reasons *how* those representations should update each other. Role-specific decoding predicts hands and the left-hand, right-hand, and two-hand object roles. On EgoHOS the method reaches 71.67 mIoU in domain and 70.75 out of domain, 6.93 points above CaRe-Ego on the out-of-domain split. Without fine-tuning it improves mini-HOI4D by 3.82 mIoU points.

1 Motivation and contributions

Learning from human demonstrations is a way to acquire manipulation skill without discovering every behavior on a robot [1]. Egocentric video shows the hands, the objects, and the way they meet [2]. The perceptual task is to segment the objects that are being manipulated now. Those objects are defined by the interaction, with the left hand, the right hand, or both, not only by how they look.

Egocentric interactive hand–object segmentation asks for the visible hands and the objects they are acting on [3]. Recent models use contact, spatial refinement, and hand–object relations [4–6]. The usual direction is hand evidence improving the object. That works when the hand is clear, and it weakens when the object hides the hand or when nearby objects compete. Figure 1 states the alternative. Object appearance, place, and contact can fill in a missing hand. Hand evidence still says which object is in play, and which hand holds it. The two sides should refine each other.

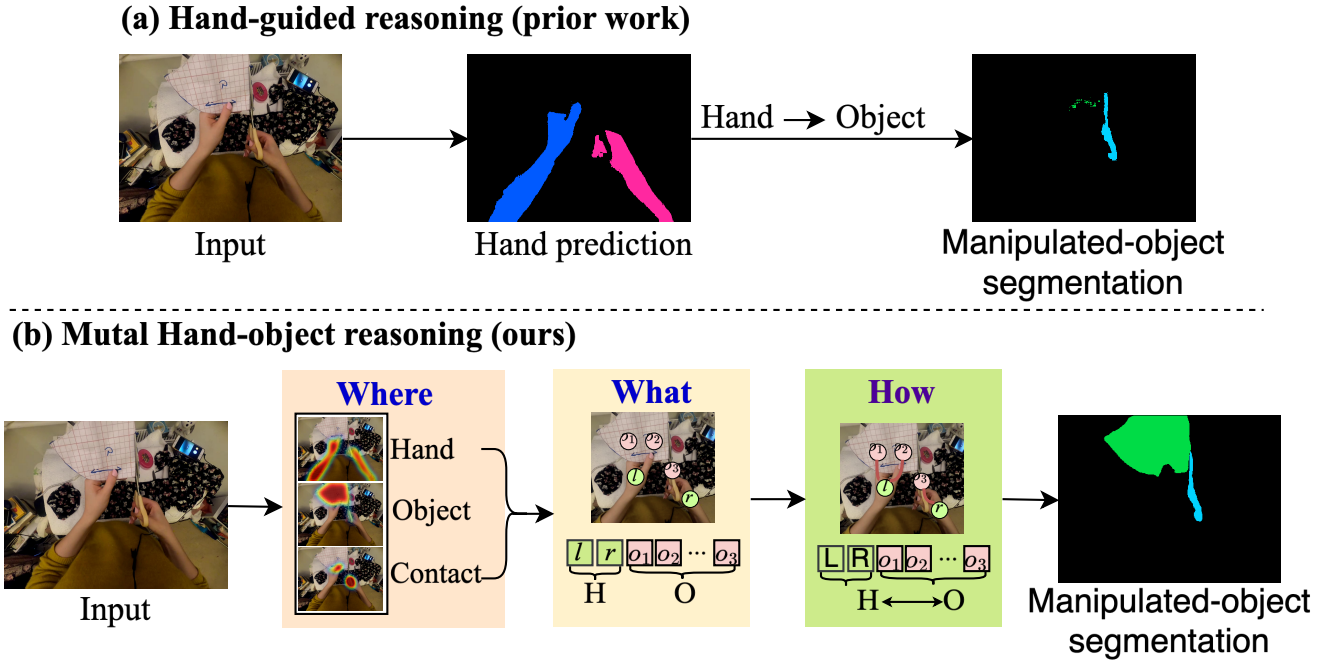


Figure 1: Motivation for mutual hand–object reasoning. Prior methods rely on hand-guided object prediction, which weakens when hand evidence is incomplete. The framework grounds where the interaction occurs, represents the participants with hand queries and object hypotheses, and lets their evidence inform one another.

The contributions are as follows.

- Interaction-grounded mutual reasoning for manipulated-object segmentation, so hand and object evidence refine one another rather than the object being only the end of a hand-guided path.
- An asymmetric representation: two fixed queries for the left and right hands, an unordered set of object hypotheses, and mutually exclusive interaction roles.
- Stronger robustness on the EgoHOS out-of-domain split and better transfer to mini-HOI4D without fine-tuning. Both directions of the exchange help, and removing the exchange hurts in-domain and out-of-domain results.

2 Method

Given an egocentric RGB image, the outputs are masks for the left and right hands and for objects manipulated by the left hand, the right hand, or both. The same physical object can change role across interactions. Hands are identity-stable, so they use two fixed queries $Q_h \in \mathbb{R}^{2\times d}$. Manipulated objects are variable and unordered, so they use N_o hypotheses Q_o, matched to annotations by the Hungarian algorithm. A visual encoder yields multi-scale features $\mathcal{F} = \{F_2, F_3, F_4, F_5\}$. Priors ground the queries, Bidirectional Hand–Object Reasoning (BHOR) refines them, and two decoders read out roles and instances (Figure 2).

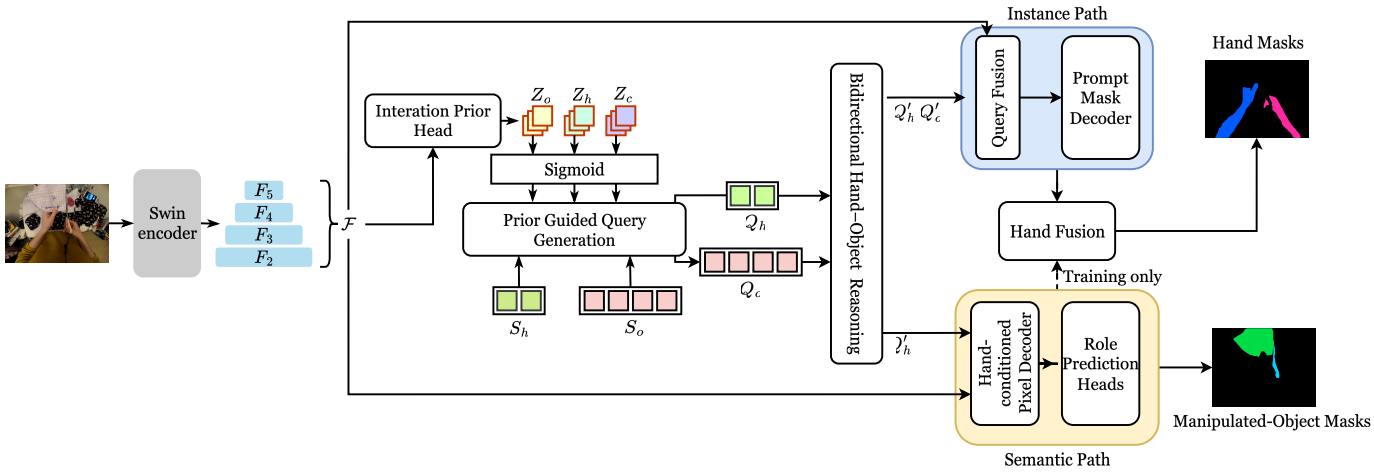


Figure 2: Overview. Hand, object, and contact priors from multi-scale features spatially ground the hand queries and object hypotheses. BHOR refines the two families. The semantic path predicts interaction roles. The instance path resolves object hypotheses and predicts the hand masks.

2.1 Where, then what

A small head on the high-resolution map F_2 predicts hand, object, and contact priors,

$$[Z_o; Z_h; Z_c] = g_{\text{prior}}(F_2), \qquad P_r = \sigma(Z_r), \quad r \in \{o, h, c\}. \tag{1}$$

These become attention biases $B_h = P_h$ and $B_o = P_o + P_c$. Hand queries are pulled toward hand evidence. Object hypotheses are pulled toward object evidence and contact. Starting from learned seeds S_r,

$$Q_r = S_r + \text{MHA}_{B_r}(S_r, F_2 + E_2, F_2), \tag{2}$$

where E_2 is positional encoding and the bias B_r is added to the attention logits. The bias guides attention and does not mask it, so a query can still read evidence outside an uncertain prior. Each family is then updated by cross-attention to a context set formed from F_4 and F_5, with separate parameters, so the left/right slots and the object set stay distinct.

2.2 How they inform each other

Each BHOR layer keeps the two structures and exchanges evidence. Object hypotheses first contextualize one another. The hand queries then read that object context, and the object hypotheses read the updated hands:

$$\begin{aligned} \tilde{Q}_h &= \text{LN}\big(Q_h + \text{MHA}(Q_h, \bar{Q}_o, \bar{Q}_o)\big), \\ \tilde{Q}_o &= \text{LN}\big(\bar{Q}_o + \text{MHA}(\bar{Q}_o, \tilde{Q}_h, \tilde{Q}_h)\big). \end{aligned} \tag{3}$$

A hand self-attention step and a feed-forward update follow. Stacking L layers exchanges evidence without making the families interchangeable. After reasoning, Q'_o is still an unordered hypothesis set. Q'_h still means left and right, and it now carries the object evidence needed to name roles.

2.3 Role-specific decoding

Multi-scale features are fused to a stride-4 map U. Each location attends to the refined hand queries, because a manipulated-object role is defined relative to the left and right hands. Object evidence is already inside Q'_h through Eq. (3). Three binary heads on this map predict the left-hand, right-hand, and two-hand object regions. A four-way softmax over background, left, right, and both is a training constraint that keeps those roles exclusive. It is not fused into the reported maps. Joint manipulation is its own class, not the overlap of two single-hand predictions.

The hypothesis set still holds entity information that a dense role map does not. An instance path decodes Q'_o and Q'_h with content and mask tokens, self-attention, and prior-biased cross-attention to multi-scale features. Each entity gets a point prompt and a mask, box, class, and contact. Object hypotheses are matched at training time. The two hand slots stay fixed. Hand masks are read from those slots. Let m_L and m_R be their mask logits and $\hat{b}$ the background logit. A learned per-channel scale $\boldsymbol{\gamma}$ gives

$$A_H = \boldsymbol{\gamma} \odot [\,\hat{b}; m_L; m_R\,], \tag{4}$$

and the hand label is the channel-wise argmax. Hands are identity-stable entities. Objects are scored by interaction role.

Training sums a prior loss, a semantic role loss, an instance loss, a distillation term, and auxiliary losses. Priors use binary cross-entropy and Dice. Semantic heads use focal and Dice losses, with a heavier weight on the single-hand maps. Matched object hypotheses receive mask, box, class, point, and contact losses. The hand slots receive mask losses and a repulsion term so both slots do not claim the same instance. Distillation uses the detached semantic role probability inside each ground-truth instance as a Dice teacher for that instance mask, so one hypothesis cannot explain every object that shares a role. Gradients from this term go only through the instance path. Intermediate BHOR layers get auxiliary object supervision, and a hand head on the dense role features is trained and then dropped at inference.

3 Results

EgoHOS has 11,743 images: 8,993 train, 1,124 validation, 1,126 in-domain test, and 500 out-of-domain test images from unseen environments [3]. The EgoHOS-trained model is also tested on 1,095 mini-HOI4D images with no fine-tuning. That split has no left-hand-object pixels, so the category is left out of its average. We report IoU for the left hand (LH), right hand (RH), left-hand object (LObj), right-hand object (RObj), and two-hand object (Both), plus their mean (mIoU).

The encoder is Swin-B, with width d=256 and 448 × 448 inputs. There are N_o=64 object hypotheses, two hand queries, and three BHOR layers with eight heads. Training runs 100 epochs with AdamW and cosine decay, peak learning rates 10^{-4} for new modules and 10^{-5} for the encoder, and batch size 32. The checkpoint is the one with the best validation mIoU. The model has 107.7M parameters and runs at 37.4 frames per second on one RTX 5000 Ada.

On the in-domain split the method reaches 71.67 mIoU, against 71.49 for CaRe-Ego, and it is ahead on all three object roles (62.82, 58.77, 55.29 against 60.07, 56.69, 54.73). CaRe-Ego remains slightly ahead on the two hands. The gap opens out of domain (Table 1). Out-of-domain mIoU is 70.75, which is 6.93 points above CaRe-Ego (63.82). The object gains are 14.06 on LObj, 18.78 on RObj, and 5.35 on Both. From in-domain to out-of-domain, mIoU falls by 0.92 points, against 7.67 for CaRe-Ego.

Table 1: EgoHOS IoU (%). ID and OOD are in-domain and out-of-domain mIoU. Object columns are out-of-domain IoU. Seq.$^\flat$ is the strongest sequential EgoHOS variant.

Method	ID	OOD	LObj	RObj	Both
Mask2Former	64.88	63.68	53.41	44.90	35.61
Seq.$^\flat$	67.17	55.26	46.93	26.40	42.38
CaRe-Ego	71.49	63.82	51.56	36.80	41.84
Ours	**71.67**	**70.75**	**65.62**	**55.58**	**47.19**

Direct transfer to mini-HOI4D reaches 65.77 mIoU, 3.82 points above CaRe-Ego (61.95). Left-hand IoU rises from 70.39 to 80.38 and two-hand object IoU from 60.08 to 65.50. Right-hand object IoU is slightly lower (27.23 against 27.56).

Table 2: Ablation on EgoHOS, mIoU (%). Obj. is the mean of LObj, RObj, and Both.

	In-domain		Out-of-domain	
Setting	Obj.	mIoU	Obj.	mIoU
Full model	**58.96**	**71.67**	**56.13**	**70.75**
No interaction priors	58.21	71.04	55.01	69.74
No object evidence	54.94	69.29	50.18	67.21
No $O \to H$	55.34	69.51	53.06	68.91
No $H \to O$	56.42	70.14	54.68	69.86
No BHOR	50.63	63.92	47.27	64.03

Table 2 separates the pieces. Dropping the interaction priors costs about one mIoU point. Removing object evidence costs 4.02 object IoU in domain and 5.95 out of domain. Cutting only the object-to-hand path, or only the hand-to-object path, hurts both splits, and the object-to-hand cut is the larger of the two out of domain. Removing BHOR entirely drops mIoU from 71.67 to 63.92 in domain and from 70.75 to 64.03 out of domain, and it is also the change that most reduces hand IoU. Both directions of the exchange contribute. Object evidence matters through the path into the hand queries, which is what the dense role decoder reads.

4 Conclusion

Hands and manipulated objects are mutually informative, and they are not the same kind of participant. The framework grounds where the interaction is, represents the two hands with fixed queries and the objects with hypotheses, lets BHOR exchange evidence, and decodes roles and instances separately. On EgoHOS the in-domain gain over a strong hand-guided baseline is small, and the out-of-domain gain is large. The same model transfers to mini-HOI4D without fine-tuning. Mutual interaction modeling is a robust way to see which object is being manipulated in egocentric video, and a perceptual front end for learning from demonstrations.

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